Hot-dry-rock energy

Natural rock fractures that bring hot water to the surface from deep underground have been described as the “low-hanging fruit” of geothermal energy because they are easy to exploit. However, they are rare in most parts of the world. The vast majority of geothermal energy locked beneath Earth’s surface is in dry, nonporous rock.

The enhanced geothermal system (EGS), a similar process to fracking for natural gas and oil, aims to overcome this problem by fracturing rock strata and injecting water into it at a great depth. The water is heated by contact with the rock, then returns to the surface through production wells. Depending on the economic limits of drill depth, the technology might be feasible across many parts of the world, but there are risks. Like fracking, EGS can cause small earth tremors, so it should not be conducted near populated areas or power stations.

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